Answer **all** the questions in the spaces provided.

1 Fig. 1.1 shows block A of mass 1.5 kg held against a spring with a force *F*. The spring is compressed by 2.0 cm.

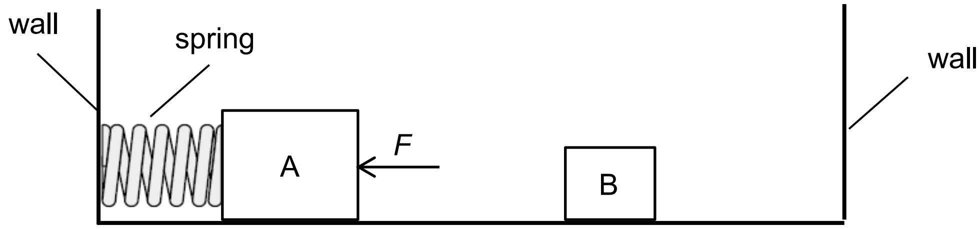


Fig. 1.1

The force *F* is then removed and the block A accelerates to the right before losing contact with the spring with a speed of 0.50 m s^{-1} as shown in Fig. 1.2. Block A collides elastically head-on with block B. The mass of block B is 0.50 kg.

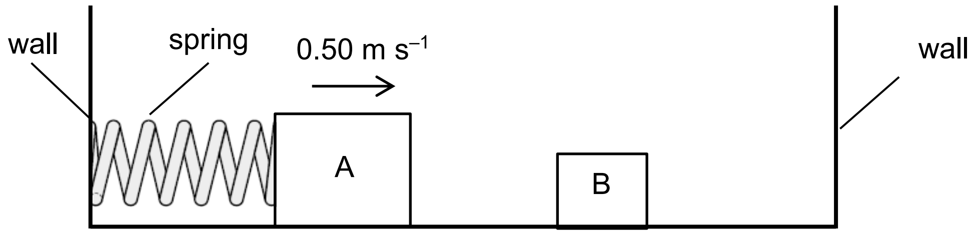


Fig. 1.2

Air resistance and frictional forces are negligible.

(a) Determine the speed of block B after the collision with block A.

final speed of block B = ... m s^{-1} [3]

Solution:

Take vectors to the right as positive.
By Conservation of Momentum,

$$m_A u + 0 = m_A v_1 + m_B v_2$$

$$1.5(0.50) = (1.5)v_1 + (0.50)v_2 \qquad (1)$$

$$0.75 = 1.5v_1 + 0.50v_2$$ 1

For head-on, elastic collision,
Relative speed of approach = Relative speed of separation

$$u - 0 = v_2 - v_1$$

$$0.5 = v_2 - v_1 \qquad (2)$$

$$v_1 = v_2 - 0.5$$ 1

Sub (2) into (1)

$$0.75 = 1.5(v_2 - 0.5) + 0.50v_2$$

$$v_2 = 0.75 \text{ m s}^{-1}$$ 1

(b) Fig 1.3 shows the variation with time of the force acting on block A during the collision with block B.

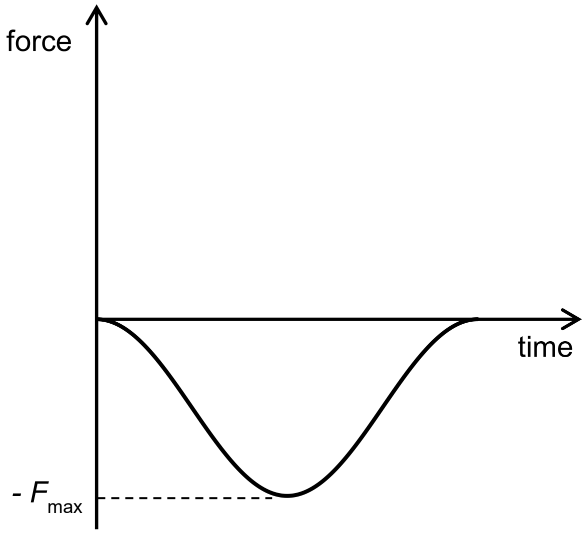
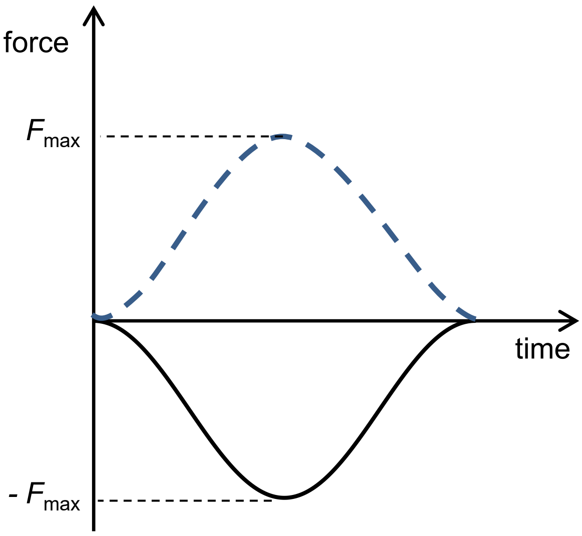


Fig. 1.3

(i) Sketch on Fig. 1.3, the corresponding graph to show how the force on block B varies with time during the collision between block A and block B.

[1]

Solution:



1

Marking points:

- Equal in magnitude and opposite in direction at every instant in time

		(ii)	Explain how your graph shows that the total momentum of the blocks remains unchanged during the collision.	
			……………………………………………………………………………………………………… ……………………………………………………………………………………………………… ………………………………………………………………………………………………………	
			………………………………………………………………………………………………………	[2]
			Solution: Area under the force-time graph represents the change in momentum of a body. The area under the graphs of A and B are equal in magnitude but opposite in direction. This implies that the change in momentum of one block is equal and opposite to the change in momentum of the other so that total momentum is conserved.	1 1
	(c)	Block B hits the wall elastically, rebounds and collides with block A. Block A then moves and compresses the spring. State, with a reason, whether the maximum compression of the spring will be greater than, less than, or equal to 2.0 cm.		
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		………………………………………………………………………………………………………		[2]
		Solution: Even though total kinetic energy is conserved, the kinetic energy of A is now less than initially as some of its kinetic energy has been transferred to B. By conservation of energy, energy transferred to the spring will therefore be less and the compression of the spring will be less than 2.0 cm.		1 1

2	A block of mass $2m$ initially rests on a track at the bottom of the circular, vertical loop of radius r as